Week 2 — Core Module Construction (Days 8–14)

Week 1 delivered curated datasets, a baseline VQC, and a feature-selection literature study. Week 2 turns each team into a builder of one QS-Net workstream. Nobody waits: Team A builds the partitions the conformal guarantee needs, Team B implements the margin-aware training that shapes the Hilbert-space geometry, and Team C implements and benchmarks the optimisation-based selectors.

Team A — Data Partitions, Adversarial/Zero-Day Sets & Classical Baselines

Members: Interns 1, 2, 3 (Amon Koike, Mohammed Owais, Iwo Wojtakajtis)

Focus this week: *Convert the Week-1 curated datasets into the exact partitions the conformal guarantee and the disentanglement experiment require, and stand up the classical baselines.*

Day	Tasks	Deliverables
Day 8	Re-partition each curated dataset into Train / Calibration / Test / Zero-Day, holding out ≥1 entire attack class as unseen zero-day (e.g., Ransomware). Document the exact class membership of every split; the calibration set must contain KNOWN classes only.	Partition specification table (per dataset) train/calibration/test/zeroday CSVs (v1)
Day 9	Verify exchangeability prerequisites for conformal: calibration and test drawn i.i.d. from the same known-class distribution. Add stratified sampling + fixed seeds so splits are reproducible.	Split-integrity report Seeded split script
Day 10	Construct the adversarial-vs-zero-day evaluation set for RQ3: take known-attack samples to be perturbed later (FGSM/PGD by Team B) plus the genuine held-out zero-day class. Define the labelling schema that keeps 'adversarial-known' and 'true-zero-day' distinguishable in evaluation.	RQ3 evaluation-set spec Adversarial-source sample pool (clean)
Day 11	Implement classical baseline #1: XGBoost detector + Isolation-Forest novelty head. Implement classical baseline #2: Autoencoder / One-Class SVM zero-day detector.	XGBoost+IF baseline AE / OC-SVM baseline
Day 12	Wire baselines to the same splits and metrics (Accuracy, F1-macro, AUROC, zero-day AUROC). Produce first baseline numbers on one dataset as the bar QS-Net must clear.	Baseline results table (dataset 1) Unified metrics module
Day 13	Extend splits + baselines to all three datasets (CIC-IoT2023, TON-IoT, BoT-IoT). Begin the statistics scaffold: 5-seed harness, mean±std, 95% CI helpers.	All-dataset baseline results Statistics harness (skeleton)
Day 14	DATA FREEZE: finalise and version all partitions; publish prototype-shaped dummy score files so Teams B and C can build against a stable interface. Hand over calibration/test/zero-day sets + baseline results to the wider project.	FROZEN dataset package (v1.0) Dummy calibration scores Handover note

Team B — Margin-Aware Quantum Training (MAQT), Prototypes & Attack Framework

Members: Interns 4, 5 (Arjun E Naik, La Wun Nannda)

Focus this week: *Turn the Week-1 baseline VQC into the geometry-shaping trainer (Algorithm 1) and build the adversarial-attack framework that later validates Proposition 2.*

Day	Tasks	Deliverables
Day 8	Refactor the Week-1 VQC onto PennyLane default.mixed so density matrices, fidelity and trace distance are exact. Add identity-block initialisation (barren-plateau safe) and gradient-variance logging.	Mixed-state VQC Gradient-variance monitor
Day 9	Implement running class-prototype computation: mean density matrix ρ_c per known class. Add Uhlmann fidelity $F(\rho,\sigma)$ and trace-distance utilities (qml.math).	Prototype module Fidelity / trace-distance utilities
Day 10	Implement the MAQT loss (Algorithm 1): $L = L_{CE} + \lambda 1 \cdot L_{intra} + \lambda 2 \cdot L_{inter}$. Unit-test each loss term on a toy 2-class problem.	MAQT loss implementation Loss unit tests
Day 11	Train MAQT end-to-end on one frozen dataset (from Team A dummy interface, swap to real on Day 14). Log intra-class and inter-class fidelity gaps (H1 diagnostic).	First MAQT training run Hilbert-geometry diagnostics v1
Day 12	Build the adversarial-attack framework: FGSM and PGD on the encoded inputs. Validate attacks on the plain VQC (pre-MAQT) to confirm they degrade accuracy.	FGSM implementation PGD implementation
Day 13	Tune $\lambda 1$, $\lambda 2$ for tight, separated prototypes; record the sensitivity (feeds the ablation later). Confirm no barren-plateau collapse (gradient variance > 1e-6).	λ-sensitivity notes Training-stability report
Day 14	Swap in the FROZEN Team-A data; retrain MAQT and emit real prototypes $\{\rho_c\}$ for one dataset. Publish prototype density matrices so Team A can calibrate conformal thresholds next week.	Real prototypes $\{\rho_c\}$ (dataset 1) MAQT model checkpoint

Team C — MO-AQPSO Implementation & Feature-Selection Benchmark (Article 2 core)

Members: Interns 6, 7 (Kevin Wong, Maral Mahmoudi Kamelabad)

Focus this week: *Move from the Week-1 designs to working BPSO / QPSO / MO-AQPSO implementations and a first comparative benchmark.*

Day	Tasks	Deliverables
Day 8	Finalise the MI and Random-Forest importance rankers from Week 1 into a reusable interface. Fix the shared evaluation classifier and metrics so all selectors are compared fairly.	MI + RF ranker module Selector evaluation harness
Day 9	Implement Binary PSO (BPSO) for feature selection end-to-end. Validate selected subset on the classical baseline classifier.	BPSO implementation BPSO preliminary subset
Day 10	Implement Quantum-Inspired PSO (QPSO). Compare BPSO vs QPSO on accuracy, #features, runtime, stability.	QPSO implementation BPSO-vs-QPSO comparison v1
Day 11	Begin implementing the MO-AQPSO framework (multi-objective: accuracy, feature count, redundancy, cost). Define the Pareto objective vector and dominance test.	MO-AQPSO skeleton Objective/dominance module
Day 12	Complete MO-AQPSO adaptive mechanism; run first optimisation to a Pareto front. Sanity-check selected subsets against MI/RF/BPSO/QPSO.	Working MO-AQPSO First Pareto front
Day 13	Benchmark all five selectors (MI, RF, BPSO, QPSO, MO-AQPSO) on dataset 1: accuracy, #features, runtime, stability. Start the Article-2 comparison table.	5-selector benchmark (dataset 1) Article-2 comparison table (draft)
Day 14	Extend the benchmark to all datasets; freeze a candidate Top-10 feature subset per dataset. Deliver candidate subsets to Team B for use in QS-Net (Article 1 link).	All-dataset selector benchmark Candidate Top-10 subsets

★ **MILESTONE —** Day 14 DATA FREEZE — Team A delivers all Train/Calibration/Test/Zero-Day partitions + classical baselines; Team B delivers first real MAQT prototypes; Team C delivers a working MO-AQPSO and candidate feature subsets.